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Selected System: Medical Imaging Assistant for Early Disease Detection

Assignment 1: AI System Specification

System Name

Thyroid Ultrasound Medical Imaging Assistant for Early Malignancy Detection - Artificial Intelligence (AI) decision support system that classifies thyroid ultrasound images as either benign or malignant and prioritizes suspicious thyroid images for clinical review.

Task 1: PEAS Specification

Performance Measures

ROC-AUC for benign/malignant discrimination

Sensitivity for detecting malignant cases

Calibration quality using Brier score/ECE

Precision for malignant predictions

F1 score

Specificity for correctly identifying benign cases

Inference speed per image

External generalization gap

Environment

Workflow for radiology and ultrasound of the thyroid using ultrasound machines, radiologists, hospital imaging systems, variable image quality, patient population, imbalance of patient types, different scanners, operator technique, and changing patient population.

Actuators

Digital outputs: malignancy probability score, benign/malignant recommendation, visual explanation/heatmap display, uncertainty warning, structured report support, triage priority flag, and case-queue prioritization.

Sensors

Thyroid ultrasound images; benign/malignant labels during development; image metadata; validation predictions for threshold selection; optional lesion masks for retrospective error analysis; external validation images; radiologist feedback for monitoring.

Task 2: Environment Classification

Partially Observable: The system mainly sees ultrasound images, while history, biopsy results, scanner variation, operator skill, and hidden clinical factors may be unavailable.

Multi-agent: It operates with radiologists, sonographers, clinicians, hospital systems, and patients rather than acting alone.
Stochastic: Ultrasound noise, disease variation, acquisition differences, and dataset shift make predictions probabilistic.
Sequential: Predictions can influence later review, follow-up imaging, biopsy decisions, and patient management.
Dynamic: New patients, workflow pressure, scanner differences, and population shifts change the environment over time.
Continuous: Images contain continuous pixel intensities and the model outputs continuous probability scores before thresholding.

Task 3: Critical Analysis

Greatest technical challenge: (partial) observability. A thyroid ultrasound alone is not enough of a picture to get a complete picture of what's going on with the patient. It may be lost for some necessary information such as symptoms, age, history, pathology results, differences of images acquired and circumstances of the lesions, in other words, the system should maintain its reliability if the hospital, dataset, scanner or patient population changes. Thus, it's not always necessary to have a fully functional diagnosis system when using an image classifier, a fully functional AI Engineer must prepare for uncertainty and domain shift and employ safe human supervision. Therefore, an AI engineer must design for uncertainty, domain shift, and safe human oversight rather than treating the image classifier as a complete diagnostic agent.

Utility function: U = 0.50(Sensitivity) + 0.25(Specificity) + 0.15(Calibration Quality) - 0.10(Computational Cost). This utility favors sensitivity because missing malignant cases is clinically risky, while still valuing specificity, reliable probability estimates, and efficient workflow. Doubling the sensitivity weight, would increase the number of doubtful cases detected by the system, which would enhance the security of screening, and lower the chances of a false negative detection. There might, however, have been more examples of non-fatal cases being discussed and more cases tested when there is a reduction in specificity. This environment could be suitable for early treatment, but this should be clinically justified.

Task 4: Structured Representation


```
{
  "system_name": "Thyroid Ultrasound Medical Imaging Assistant for Early Malignancy
Detection",
  "peas": {
    "performance_measures": [
      "roc_auc",
      "sensitivity",
      "specificity",
      "precision",
      "f1_score",
      "calibration_quality",
      "inference_speed",
      "external_generalization_gap"
```

```
    ],
    "environment": "Clinical thyroid ultrasound workflow with radiologists, sonographers, patients, ultrasound machines, hospital image systems, variable image quality, scanner differences, and changing patient populations.",
    "actuators": [
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      "diagnostic_recommendation",
      "triage_flag",
      "uncertainty_warning",
      "report_support",
      "visual_explanation",
      "queue_prioritization"
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      "image_metadata",
      "training_labels",
      "validation_predictions",
      "external_validation_images",
      "optional_lesion_masks",
      "radiologist_feedback"
    ]
  },
  "environment_classification": {
    "observability": {
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      "justification": "Images do not include all patient history, biopsy, scanner, and hidden clinical factors."
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    "agents": {
      "choice": "Multi-agent",
      "justification": "The system interacts with clinicians, radiologists, sonographers, hospital systems, and patients."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Image noise, disease variation, and dataset shift make outcomes uncertain."
    },
    "episodic_or_sequential": {
      "choice": "Sequential",
      "justification": "The prediction can influence later review, biopsy, follow-up, and treatment decisions."
    },
    "static_or_dynamic": {
      "choice": "Dynamic",
      "justification": "Patients, workload, scanners, and clinical distributions change over time."
    },
    "discrete_or_continuous": {
      "choice": "Continuous",
      "justification": "Images and probability scores are continuous before thresholding into labels."
    }
  },
  "utility_function": "U = 0.50*sensitivity + 0.25*specificity + 0.15*calibration_quality - 0.10*computational_cost"
}
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